

Initial Date: 4/2010

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Section: 10-3

Nerve Agent/Organophosphate Pesticide Exposure Treatment

Purpose: This protocol is intended for EMS personnel at all levels that have been trained in the use of these devices and authorized by the medical control authority to assess and treat patients exposed to nerve agents and organophosphate pesticides utilizing the **Duo Dote/Mark I Antidote Kits** and/or a combination of auto injectors and/or nasal sprays. Administration of non-prepackaged kits is restricted to ALS.

The following medications in this protocol are not required to be carried on EMS vehicles and may be available through special response units.

Medications/Definitions:

- A. One (1) Nerve Agent (NA) Antidote Kit – for the purpose of this protocol means either one (1) Duodote OR one (1) Mark I
 - 1. **Duodote** – a single device with 2 chambers. The front chamber contains 2.1 mg atropine, the back chamber contains 600 mg pralidoxime (2-PAM). When activated the device sequentially administers both drugs through a single needle.
 - 2. **Mark I Antidote kit**– 2 separate injectors. One containing 2mg atropine, the second containing 600 mg of pralidoxime (2-PAM).
- B. **Atropine auto injector**- a single auto-injector of atropine that comes in 3 doses: atropine 0.5 mg, atropine 1 mg, atropine 2 mg.
- C. **Midazolam auto-injector** – 20 mg midazolam per device
- D. **Midazolam nasal spray** – 5 mg per device
- E. **Diazepam auto-injector** – 10 mg per device
- F. Non prepackaged kit administration: Administer 600 mg **pralidoxime** and 2 mg of **atropine** for every one (1) NA Antidote Kit.(ALS only)

Chemical Agents

- 1. Agents of Concern
 - A. Military Nerve Agents including: Sarin (GB), Soman (GD), Tabun (GA), VX
 - B. Organophosphate Pesticides (OPP) including Glutathione, Malathion, Parathion, etc.
- 2. Detection: The presence of these agents can be detected through a variety of monitoring devices available to most hazardous materials response teams and other public safety agencies.

Patient Assessment

- 1. **SLUDGEM** Syndrome
 - A. **S** Salivation / Sweating / Seizures
 - B. **L** Lacrimation (Tearing)
 - C. **U** Urination
 - D. **D** Defecation / Diarrhea
 - E. **G** Gastric Emptying (Vomiting) / GI Upset (Cramps)

Michigan
SPECIAL OPERATIONS
NERVE AGENT/ORGANOPHOSPHATE PESTICIDE
EXPOSURE TREATMENT

Initial Date: 4/2010

Revised Date: 03/24/2023

Section: 10-3

- F. **E** Emesis
- G. **M** Muscle Twitching or Spasm
- 2. Threshold Symptoms: These are symptoms that may allow rescuers to recognize that they may have been exposed to one of these agents and include:
 - A. Dim vision
 - B. Increased tearing / drooling
 - C. Runny nose
 - D. Nausea/vomiting
 - E. Abdominal cramps
 - F. Shortness of breath

NOTE: Many of the above may also be associated with heat related illness.

- 1. Mild Symptoms and Signs:
 - A. Threshold Symptoms *plus*:
 - B. Constricted Pupils*
 - C. Muscle Twitching
 - D. Increased Tearing, Drooling, Runny Nose
 - E. Diaphoresis
- 2. Moderate Symptoms and Signs
 - A. Any or all above *plus*:
 - B. Constricted Pupils
 - C. Urinary Incontinence
 - D. Respiratory Distress with Wheezing
 - E. Severe Vomiting
- 3. Severe Signs
 - A. Any or All of Above *plus*
 - B. Constricted Pupils*
 - C. Unconsciousness
 - D. Seizures
 - E. Severe Respiratory Distress

***NOTE:** Pupil constriction is a relatively unique finding occurs early and persists after antidote treatment. The presence of constricted pupils with SLUDGEM findings indicates nerve agent / OPP toxicity. Constricted pupils may not be present with localized dermal exposure.

Personal Protection

- 1. Be Alert for secondary device in potential terrorist incident
- 2. Personal Protective Equipment (PPE)
 - A. Don appropriate PPE as directed by Incident Commander.
 - B. Minimum PPE for Non-Hot Zone (i.e., DECON Zone)
 - a. Powered Air Purifying Respirator or Air Purifying Respiratory with proper filter
 - b. Chemical resistant suit with boots
 - c. Double chemical resistant gloves (butyl or nitrile)
 - d. Duct tape glove suit interface and other vulnerable areas
- 3. Assure EMS personnel are operating outside of Hot Zone
- 4. Avoid contact with vomit if ingestion suspected – off gassing possible

Michigan
SPECIAL OPERATIONS
NERVE AGENT/ORGANOPHOSPHATE PESTICIDE
EXPOSURE TREATMENT

Initial Date: 4/2010

Revised Date: 03/24/2023

Section: 10-3

5. Assure patients are adequately decontaminated *prior* to transport
 - A. Removal of outer clothing provides significant decontamination
 - B. Clothing should be removed before transport
 - C. DO NOT transport clothing with patient
6. Alert hospital(s) as early as possible

Patient Management (After Evacuation and Decontamination)

1. Evaluate and maintain the airway, provide oxygenation and support ventilation as needed.
2. NOTE: Anticipate need for extensive suctioning
3. Administer appropriate number of NA Antidote kits (**Duo Dote OR Mark I**) kits per Chart A. below.
 - A. NOTE: For NA kit administration only:
 - i. Adult is > 8 years of age
 - ii. Pediatrics is $\leq$ 8 years of age
 -  B. NOTE: Medical Control contact is required prior to administration for:
 - i. Patients that meet self-administration criteria
 - ii. Patients that meet mild symptoms and signs criteria in chart below:

Michigan
SPECIAL OPERATIONS
NERVE AGENT/ORGANOPHOSPHATE PESTICIDE
EXPOSURE TREATMENT

Initial Date: 4/2010

Revised Date: 03/24/2023

Section: 10-3

	Clinical Findings	Signs/Symptoms	Required Conditions	NA Kits To Be Delivered
SELF-RESCUE	Threshold Symptoms	• Dim vision • Increased tearing • Runny nose • Nausea/vomiting • Abdominal cramps • Shortness of breath	Threshold Symptoms <i>-and-</i> Positive evidence of nerve agent or OPP on site  Medical Control Order	1 NA Kit (self-rescue)
	Mild Symptoms and Signs	• Increased tearing • Increased salivation • Dim Vision • Runny nose • Sweating • Nausea/vomiting • Abdominal cramps • Diarrhea	 Medical Control Order	1 NA Kit
	Moderate Symptoms and Signs	• Constricted pupils • Difficulty breathing • Severe vomiting	Constricted Pupils	2 NA Kits
	Severe Signs	• Constricted pupils • Unconsciousness • Seizures • Severe difficulty breathing	Constricted Pupils	3 NA Kits (If 3 NA Kits are used, administer 1 st dose of available benzodiazepine)

Michigan
SPECIAL OPERATIONS
NERVE AGENT/ORGANOPHOSPHATE PESTICIDE
EXPOSURE TREATMENT

Initial Date: 4/2010

Revised Date: 03/24/2023

Section: 10-3

Clinical Findings	Signs/Symptoms	Required Conditions	NA Kits To Be Delivered
PEDIATRIC < 8 years of age	Pediatric Patient with Non-Severe Signs/Symptoms	Threshold Symptoms -and- Positive evidence of nerve agent or OPP on site  Medical Control Order	1 NA Kit
	Pediatric Patient with Severe Signs/Symptoms	- Constricted pupils - Unconsciousness - Seizures - Severe difficulty breathing Severe breathing difficulty Weakness	1 NA Kit

-  4. Establish vascular access per **Vascular Access and IV Fluid Therapy-Procedure Protocol** when feasible, do NOT delay medication administration
-  5. If NA Antidote kit is not available:
 - A. Administer **atropine auto injector** 2 mg IM for every 1 NA Kit- that is required.
 -  B. Administer atropine 2 mg IV/IM for every 1 NA Kit that is required
 - C. Administer 600 mg pralidoxime IV/IM for every 1 NA Kit that is required (when available)
-  6. Treat seizures
 - A. Adult (> 14 years of age)
 - a. Administer **midazolam** 10 mg IM or 5 mg IN
 1. If available, midazolam auto-injector or midazolam nasal spray may be utilized, ensure total dose (regardless of dosage per device) equals 10 mg IM or 5 mg IN.
 - OR
 - b. Administer **Valium (diazepam)** auto-injector.
 -  B. Pediatrics (≤ 14 years of age)
 - a. Administer **midazolam** 0.1 mg/kg IM (maximum individual dose 10 mg) or 5 mg IV/IO/or IN
 - OR
 1. If available, **diazepam auto-injector** or **diazepam nasal spray** may be utilized, ensure total dose (regardless of dosage per device) does not exceed 10 mg IM or 5 mg IN.

Initial Date: 4/2010

Revised Date: 03/24/2023

Section: 10-3



7. Monitor EKG

8. For continued secretions, contact Medical Control and administer additional **atropine** per orders.

A. Adults (> 14 years of age) **atropine** 2 mg IV/IM



B. Pediatrics ($\leq$ 14 years of age) **atropine** 0.05 mg/kg IV/IM

Nerve Agent/Organophosphate Antidotes/Countermeasures

Weight	Age	Duodote ¹ Mod-Severe Sxs	Atropen ² (1 mg) Mod- Severe Sxs	Atropine Dose (0.1 mg/kg) IM/IV/IO	Atropine Vial ² (1 mg/mL)	Cardiac Atropine ^{2,3} (1 mg/10 mL)	Midazolam ⁴ (10 mg/2 mL) IM/IV/IO
3-5 kg (6-11 lbs)	0-2 months	1	1	0.4 mg	0.4 mL	4 mL	0.1 mL
6-7 kg (13-16 lbs)	3-6 months	1	1	0.7 mg	0.7 mL	7 mL	0.2 mL
8-9 kg (17-20 lbs)	7-10 months	1	1	0.9 mg	0.9 mL	9 mL	0.2 mL
10-11 (21-25 lbs)	11-18 months	1	1	1 mg	1 mL	10 mL	0.2 mL
12-14 kg (26-31 lbs)	19-35 months	1	2	1.3 mg	1.3 mL	13 mL	0.25 mL
15-18 kg (32-40 lbs)	3-4 years	1	2	1.6 mg	1.6 mL	16 mL	0.3 mL
19-23 kg (41-51)	5-6 years	1	2	2 mg	2 mL	20 mL	0.4 mL
24-29 kg (52-64)	7-9 years	2	3	2.6 mg	2.6 mL	26 mL	0.5 mL
30-36 kg (65-79 lbs)	10-14 years	2	3	3.3 mg	3.3 mL	33 mL	0.6 mL
Adult	>14 years	2 to 3	4 to 6	4 to 6 mg	4 to 6 mL	40-60 mL	2 mL

¹Preferred initial autoinjector, ²May Repeat atropine every 5 minutes until airway secretions decrease (6 mg maximum), ³Not available in MEDDRUN, ⁴Patients with severe symptoms should receive midazolam even if not obviously seizing

Medication Protocols

Atropine

Midazolam

Nerve Agent Antidote Kit

Pralidoxime